

Inter-City Networking Solution - Network Design for F-Corp.

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Abstract

This report presents the design, configuration, and evaluation of a networking solution for a small to medium-sized enterprise (SME). The proposed network includes fail over systems to minimize single points of failure, redundancy for reliability, and security features—such as Secure Shell (SSH) and guest wireless segmentation—to mitigate network attacks. Key components include routers, wireless segmentation, DHCP, HSRP, and SSH-based access control, to help the SME reach its business targets and operational requirements.

The evaluation assesses cost, efficiency, scalability, and security, focusing on overall performance and reliability. Allowing the business to maintain its confidentiality, integrity and availability. Results show the design meets the SME's requirements for a robust, secure, and functional network. Supplementary screenshots and a video taken on Cisco Packet Tracer showcase connectivity tests and network configurations to provide additional validation for the proposed network design.

Introduction

F-Corp, a successful technology and digital services provider, has recently purchased new office spaces in two locations: Manchester and London. This acquisition marks the beginning of their venture into accounting software development. They have requested a network suitable for both the accounting department and the newly hired software development team. The proposal includes a total of 20 hosts across two offices in each location. This setup would support 40 users in total, evenly split between the accounting and software teams.

As the Senior Network Architect employed by F-Corp, the proposed topology, configuration and deployment of the network was my responsibility. To accommodate the request, a network topology was created using Cisco Packet Tracer, a comprehensive networking simulation software tool (Cisco, n.d.), which aims to provide a working overview of the proposed network and its ultimate goal. The discussion of which is available in the following sections of this document; Network Design and Network Evaluation and Discussion. The relevant reasoning for the proposed technologies is also showcased, identifying areas for improvement should F-Corp wish to scale this department in future.

Aims and Objectives

The primary objective of this project was to provide F-Corp with a functioning, scalable and efficient network solution. Several important factors were vital in ensuring these aims were met:

- CIDR Addressing Scheme.
- RIP Routing.
- Scalable and Efficient Subnetting.
- Single Point of Failure (SPOF) Redundancies.
- Secure Shell (SSH).
- Dynamic Host Control Protocol (DHCP).
- L3 Router and L2 Switch Hostname and Password Configuration.
- Wireless Connectivity for Additional Devices ie. Laptops and mobile phones.

These objectives led to a network that is resilient to outages, suitable for expansion and has multiple security features to mitigate external and internal disruption.

Network Design

Network Plan

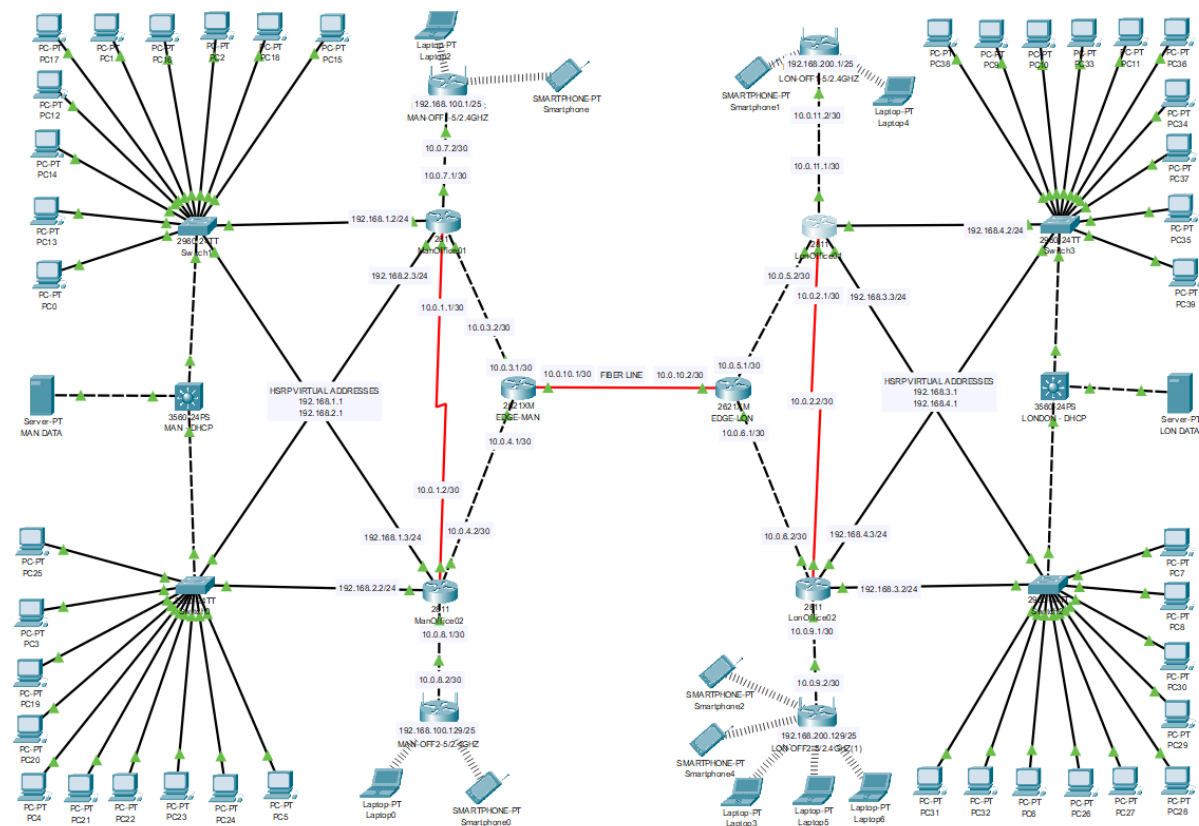


Figure 1: Full network topology with subnetting scheme.

Hardware Configuration

To address the location requirements of the proposed business plan, it was integral that each location had enough networking components to meet the required specification of users. For this, a combination of two 2811 and one 2621XM Cisco routers per building yielded the best results. This resulted in each office space having its own dedicated subnet and default gateway for network traffic. This also allowed different departments to be segmented both physically and logically using the subnetting scheme. This reduces the amount of network traffic traveling between routers both internally and externally, as each subnet has a designated router to route traffic across the network. A combination of Layer 2 (L2) Switches and cheaper home routers make up the components for each office floor.

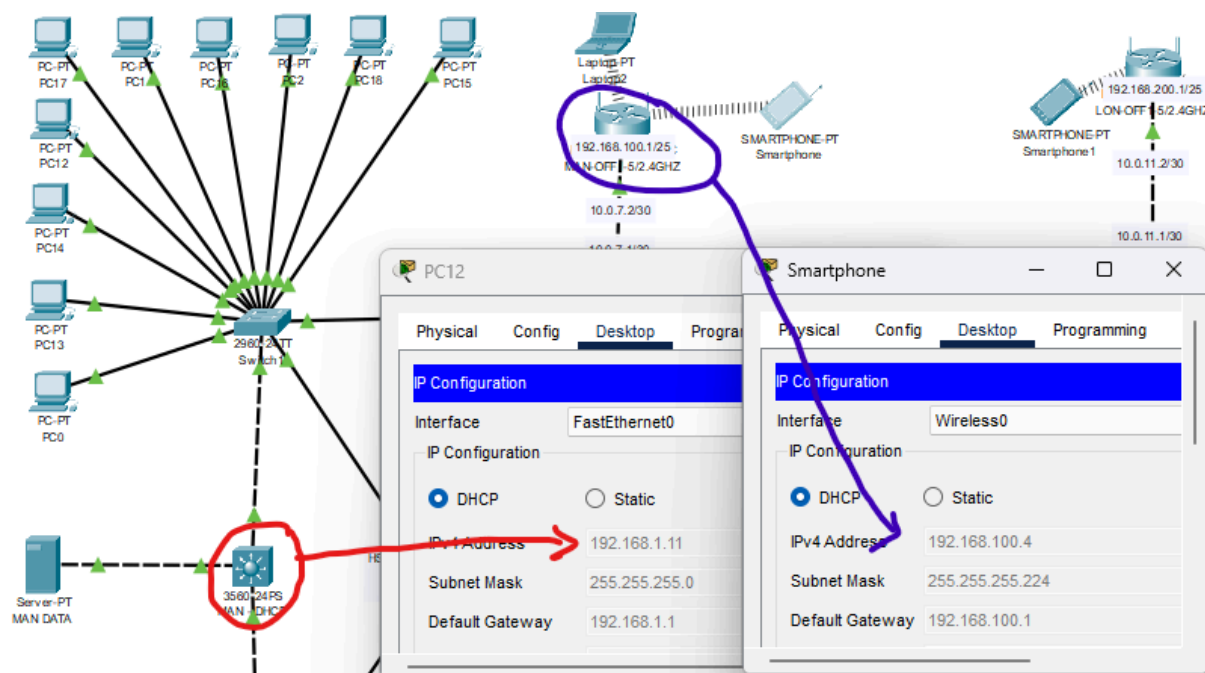


Figure 2: Wireless connectivity and DHCP assigned IP addresses.

The primary office router handles both the wired Ethernet traffic and forwards the traffic from devices connected to the Wireless Access Point (WAP). Additionally a Layer 3 or Multilayer Switch connects between the subnets of each office floor. This switch acts as the Dynamic Host Control Protocol (DHCP) Server for devices on either subnets. DHCP allows for automatic and centralised assignment of IP addresses within a network with RFCs 2131 and 2132 defining DHCP as an Internet Engineering Task Force (IETF) standard (Microsoft, n.d.) This allows new wired to request an IP address upon connecting to the switch and also handles revoking of IP addresses should a host disconnect. The wireless DHCP for the guest wireless network is handled by the wireless routers, each office location is given a dedicated range of IP addresses for guest devices, this provides additional separation for security reasons.

This L3 switch also routes traffic to and from the data server, which handles internal data storage for the building and both of the different offices. This allows for the easy interchange of data and stored information between departments.

A combination of copper straight through and cross-over cables were used throughout the network design, with crossover cables utilised between networking devices and straight

through for wired hosts. A singular serial cable was used between the two primary office routers as a final piece of redundancy and this is reflected in the HSRP configuration.

Subnetting Scheme

To establish a scalable subnetting scheme, each router was assigned a **10.x.x.x** address. The serial link between internal routers exists on a subnet with no additional networking devices. As well as Point-to-Point connections between WAP and between the edge routers of each building using the Fiber transfer medium. This was achieved by subnetting the network using a **/30** or **255.255.255.252** subnet mask. Providing a total of four addresses, one network, one broadcast and two usable IPs. Originally, a **/31** subnet was chosen as newer networking devices have implemented change to remove the redundant broadcast address on Point-to-Point networks (NetSec Cloud, 2024), but this wasn't achievable due to hardware restrictions within Packet Tracer. With this subnetting scheme, more routers can be assigned within the **10.x.x.x** address range.

Within the smaller subnets for each office **/24** or **255.255.255.0** subnetting scheme was chosen. Each office has been assigned a network range consisting of 253 hosts. Within each subnet, a range of 8 addresses have been marked as non-assignable by the DHCP server as they are allocated for networking devices only. This also facilitates the implementation of HSRP between internal routers to reduce SPOFs. This subnetting scheme allows for many more hosts to be added to the network, should the network have to be scaled upwards. Using an incremental network address scheme means more networks can be added to each router with ease. With the Manchester location taking **192.168.1.0/24** and **192.168.2.0/24** and London taking **192.168.3.0/24** and **192.168.4.0/24** respectively.

Network Configuration

Dynamic Host Control Protocol (DHCP)

As devices enter the network, it is important that they are assigned an IP address so they can communicate across the network using L3 protocols. Statically assigning IPs on a network of this size would be tedious and impractical. To address this, each building has a L3 Switch operating as a DHCP server. Any new wired devices that connect to the network are assigned an IP address associated with the subnet in which they are connecting to. Additionally, DHCP has been configured on the WAPs to provide IP addresses to wireless devices so they can access secure resources. With guest devices existing on a separate subnet so they can be isolated.

```

Switch#show ip dhcp pool

Pool LonOffice01 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 12
  Excluded addresses : 2
  Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
192.168.3.1        192.168.3.1 - 192.168.3.254  12 / 2 / 254

Pool LonOffice02 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 15
  Excluded addresses : 2
  Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
192.168.4.1        192.168.4.1 - 192.168.4.254  15 / 2 / 254
Switch#

Switch#show ip dhcp pool

Pool ManOffice01 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 11
  Excluded addresses : 2
  Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
192.168.1.1        192.168.1.1 - 192.168.1.254  11 / 2 / 254

Pool ManOffice02 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 14
  Excluded addresses : 2
  Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
192.168.2.1        192.168.2.1 - 192.168.2.254  14 / 2 / 254
Switch#

```

Figure 3 & 4: Manchester and London DHCP Pools for each of the 4 offices.

Using a L3 Switch and cheap routers meant that an additional server, which is costly to purchase and operate, isn't required for the network to function. It also provides an extra level of redundancy between devices within the office building should both primary office routers go offline. In this scenario, internal data storage is still maintained but external site-to-site/internet connectivity would be lost. This process was repeated at both locations as there was no need to differ and create confusion between different buildings based on the briefing provided for the businesses functions.

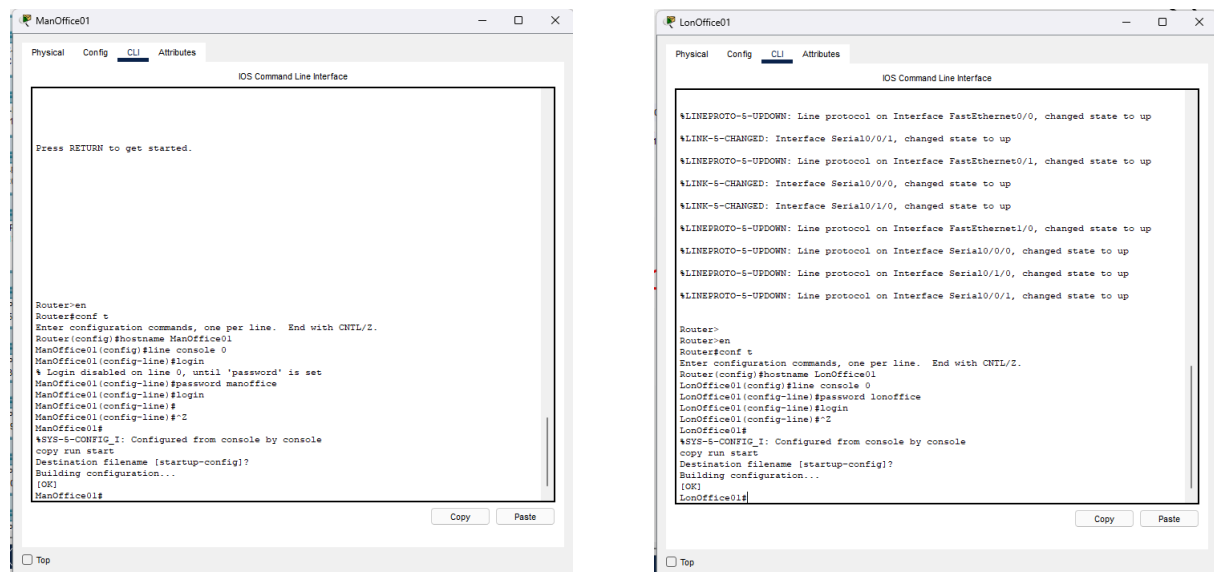
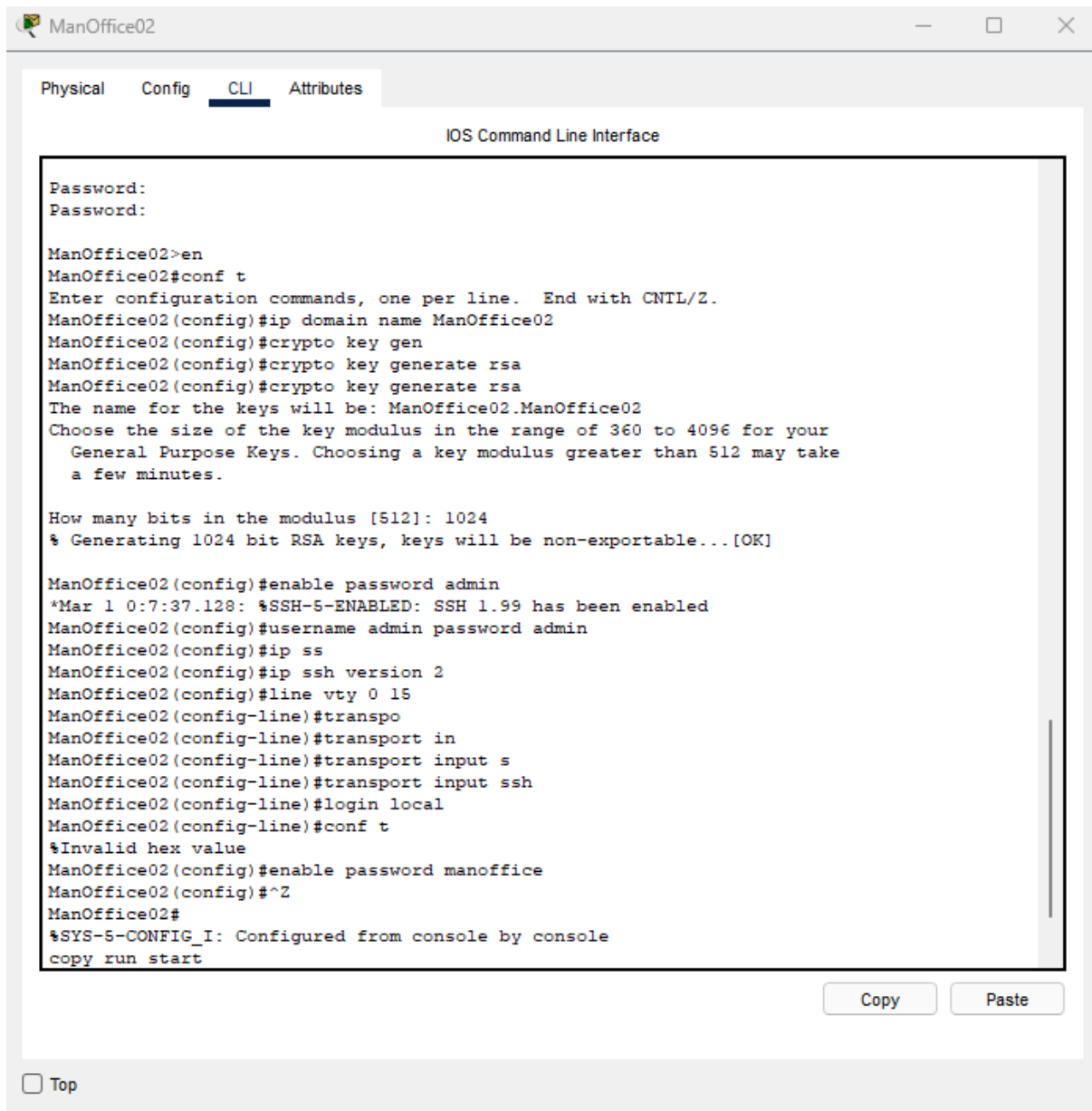


Figure 5: Configuration of hostnames and passwords for securing access to the routers console.

Secure Shell (SSH)

Secure Shell (SSH) is a networking tool that helps in managing network infrastructure and other mission-critical system components. As well as offering file transfer and other utilities remotely between hosts (Ylonen, T., 1996). After configuring hostnames and passwords for enabling and configuring the routers, SSH provides remote access to these routers should they

need reconfiguring in future. After enabling SSH v2 on each of the routers, a basic username and password for an administrator is present, with this being changeable to increase security. Not only this but SSH keys, a public key cryptography for authenticating hosts and users (Ylonen, T. 1996) was generated using the RSA algorithm, a cryptographic system based on prime numbers (Okta, n.d.). This allows administrators and trusted users to remotely access system resources securely over the network, a necessary function in a modern network.



```
ManOffice02>en
ManOffice02#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ManOffice02(config)#ip domain name ManOffice02
ManOffice02(config)#crypto key gen
ManOffice02(config)#crypto key generate rsa
ManOffice02(config)#crypto key generate rsa
The name for the keys will be: ManOffice02.ManOffice02
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

ManOffice02(config)#enable password admin
*Mar 1 0:7:37.128: %SSH-5-ENABLED: SSH 1.99 has been enabled
ManOffice02(config)#username admin password admin
ManOffice02(config)#ip ss
ManOffice02(config)#ip ssh version 2
ManOffice02(config)#line vty 0 15
ManOffice02(config-line)#transpo
ManOffice02(config-line)#transport in
ManOffice02(config-line)#transport input s
ManOffice02(config-line)#transport input ssh
ManOffice02(config-line)#login local
ManOffice02(config-line)#conf t
%Invalid hex value
ManOffice02(config)#enable password manoffice
ManOffice02(config)#^Z
ManOffice02#
%SYS-5-CONFIG_I: Configured from console by console
copy run start
```

Figure 6: Generating RSA SSH Keys and setting default credentials for Manchester Office 2.

Routing Information Protocol (RIP)

To improve routing efficiency within the network, the Routing Information Protocol (RIP) was implemented. RIP uses hop count as the metric to rate the value of different routes (Cisco,

n.d.b), this makes RIP suitable for smaller networks like the one proposed for F-Corp. Other distance vector protocols such as Interior Gateway Routing Protocol (IGRP) could have been implemented as it offers faster convergence and increased scalability (GeeksForGeeks, 2024). However, for a network of F-Corps size and required usage, the implementation of IGRP would be exorbitant. RIP is highly compatible with the LAN setup in our network design, providing simple installation and easy troubleshooting at the trade offs of lower hop count and less convergence speed, with a maximum hop count of 15(Hendrik, 1988).

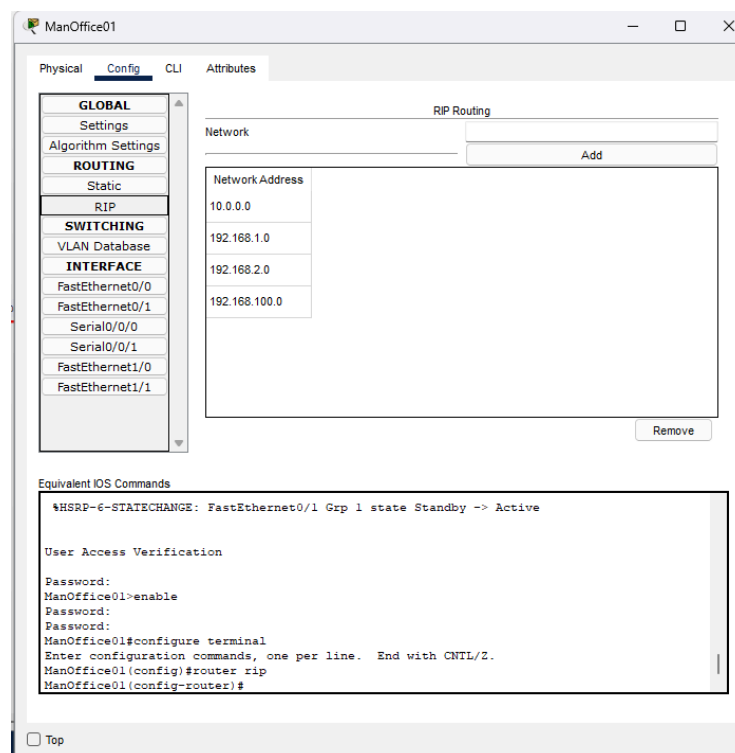


Figure 7: RIP configuration for the internal office based router in Manchester Office 1.

Hot Standby Routing Protocol (HSRP)

To reduce the number of Single Point of Failures (SPOF) within the network configuration, the HSRP protocol was used to help reroute traffic between subnets should one of the offices main routers lose connectivity. Additional edge routers with HSRP was considered but would have been too costly to implement as an additional two more expensive routers would be required across the network. In HSRP, a set of routers work in conjunction with each other to present the illusion of a single virtual router. (Cisco Systems, 1998). This was used to create effectively two virtual routers for each subnet. Each router has an interface that links to the other offices subnet and as such acts as the primary and standby IP in the case of failure.

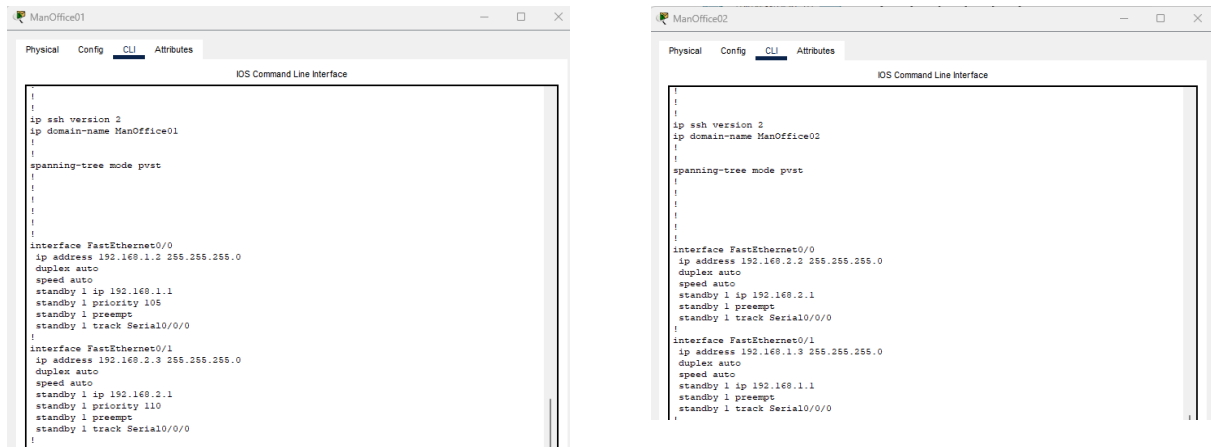


Figure 8: HSRP configuration on routers within the office building to reduce SPOFs.

Network Evaluation and Discussion

The proposed network design implements numerous features to increase security, availability and scalability. For security, encrypted protocols such as SSH for remote communication have been utilised with different credentials for trusted users such as system administrators. This means unauthorised users are unable to access router configurations without the necessary permissions to do so. Not only this but each router is configured with different hostnames and passwords that correlate with the network and location they belong to, providing additional security requirements for router configuration.

Although the amount of network devices required to achieve this network design may seem more than what is required to make the network function, the increase in devices also provides an increase in both network performance and redundancy from potential failure. Each router is interconnected with its opposite in the same building, reducing the chance of a Single Point of Failure (SPOF) outage. If one router loses connection the secondary office router will continue to route traffic from both subnets at a slight performance cost. Other issues such as DHCP failure is also accounted for using HSRP. This means the network will continue to function to give time for diagnosis and resolution of issues. The use of a multilayer switch also provides an additional route between internal building hosts, meaning some level of communication will still be possible if all external routers fail. Although, the use of DHCP on this multilayer switch could propose a failure if this switch fails, the data server could then be used as a backup DHCP server to maintain functionality in this scenario. Network cables can be costly however, with the number of Ethernet cables required to create this setup it may create some issues if expanding on a less than expected budget.

In the case that F-Corp does wish to expand its software development division to produce new software solutions outside the accountancy field, this network can be scaled easily at some economic cost. New offices within the same building can be connected to the existing infrastructure using Point to Point serial connections existing on a /30 subnet. Although there may be a cost implication when using more Ethernet cables, should the network continually

grow. Additionally there is a large amount of IP address space for each office to be allocated, the expansion would have to include several new locations with many offices before issues of scalability are encountered. With the use of DHCP, new hosts are also easily allocated IP addresses within the desired subnet, there is currently enough space for 253 hosts to exist at one time, with only 10 wired hosts within the office, many wireless devices such as laptops, phones and tablets can be connected and disconnected without fear of IP exhaustion. Although in larger business scenarios, many devices for general building maintenance and safety require wireless connection and as such could cause issues if an entire building was bought for F-Corps hypothetical expansion.

Other issues may arise with this network solution, as currently the subnets are logically and physically segmented using this subnetting scheme, there is little software based security between these networks. No access rules have been established between the different subnets, allowing any user to communicate across the network to any department, this could create security issues. To reduce the risk, departments may have different rules on the areas of the networks they are allowed to communicate depending on their authorisation and job requirement or specification. This would need to be brought about and detailed within a form of acceptable use policy.

Conclusion

The designed network solution fulfills the requirements of F-Corps proposed new software development department, through the use of modern networking hardware and logical addressing. The topology of this network provides a simple yet robust solution that can withstand a range of different network failures, such as individual host outages, router issues and DHCP failures. In addition, the network can be scaled in each existing location to provide an increased number of wireless and wired hosts, additionally further expansion is made possible through the use of smart addressing schemes and existing ports with even further development available if multiple new sites were to exist. Some issues exist with potential security risks in the current configuration which can be mitigated through appropriate access controls. Networks speeds could also be affected in certain scenarios with the use of slower serial cable mediums between office routers, this could be reformatted if an issue became present.

References

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Cisco. (n.d.b). *Routing Information Protocol*. Cisco Configuration Guide. Available at: https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst2960xr/software/15-2_5_e/configuration_guide/b_1525e_consolidated_2960xr_cg/routing_information_protocol.html [Accessed 21 Nov. 2024]

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C. Hendrik (June, 1988). "[RFC 1058 Section 2.2](https://www.rfc-editor.org/rfc/rfc1058)". *Routing Information Protocol*. The Internet Society. doi:10.17487/RFC1058.

Cisco Systems (1998). *Cisco Hot Standby Router Protocol (HSRP)*. RFC 2281. Available at: <https://datatracker.ietf.org/doc/html/rfc2281> [Accessed: 17 June 2024].

Appendices

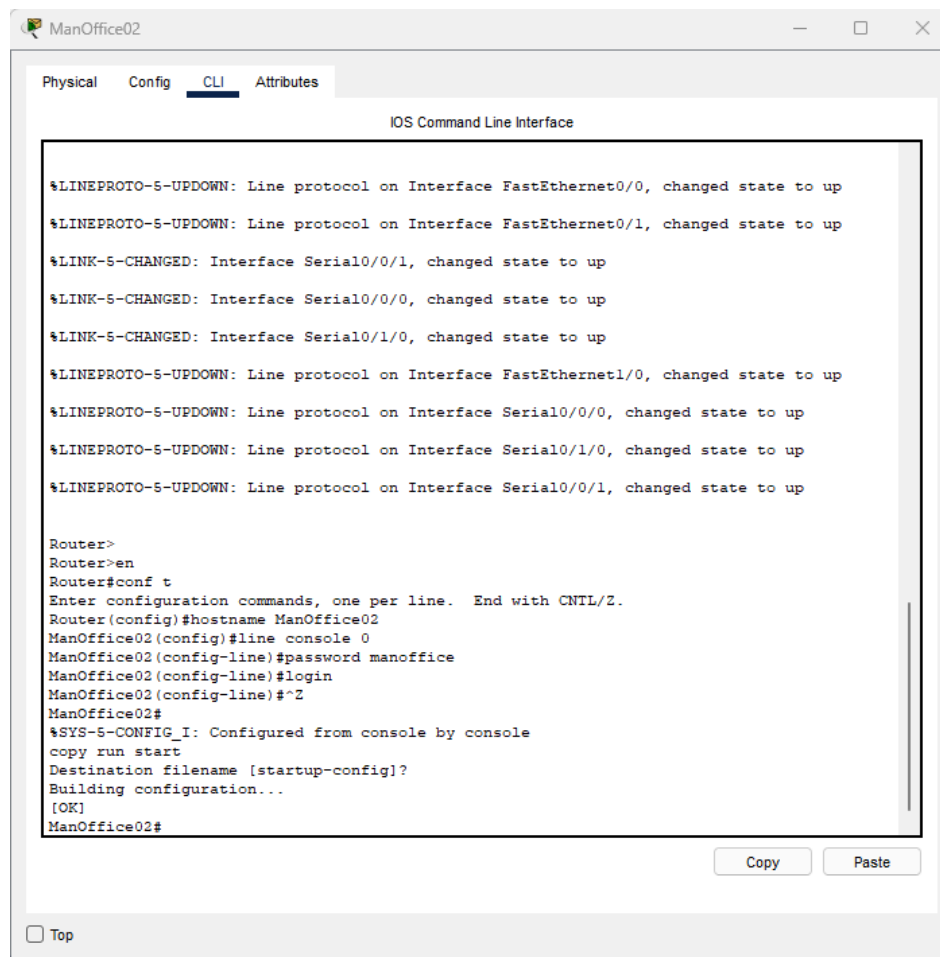
```
IOS Command Line Interface

Press RETURN to get started!

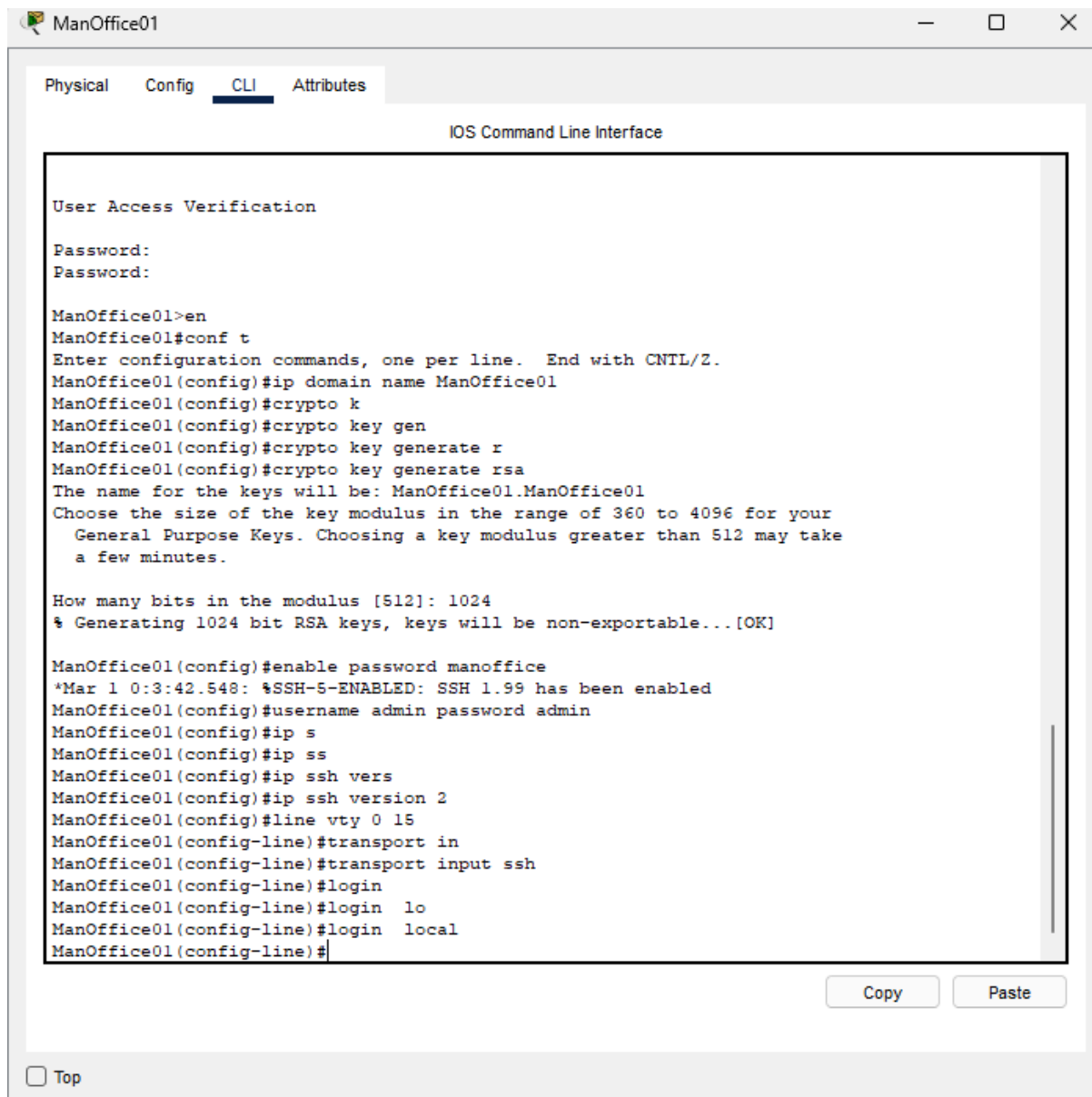
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname LonOffice02
LonOffice02(config)#line console 0
LonOffice02(config-line)#password lonoffice
LonOffice02(config-line)#login
LonOffice02(config-line)#^Z
LonOffice02#
%SYS-5-CONFIG_I: Configured from console by console
copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
LonOffice02#
```

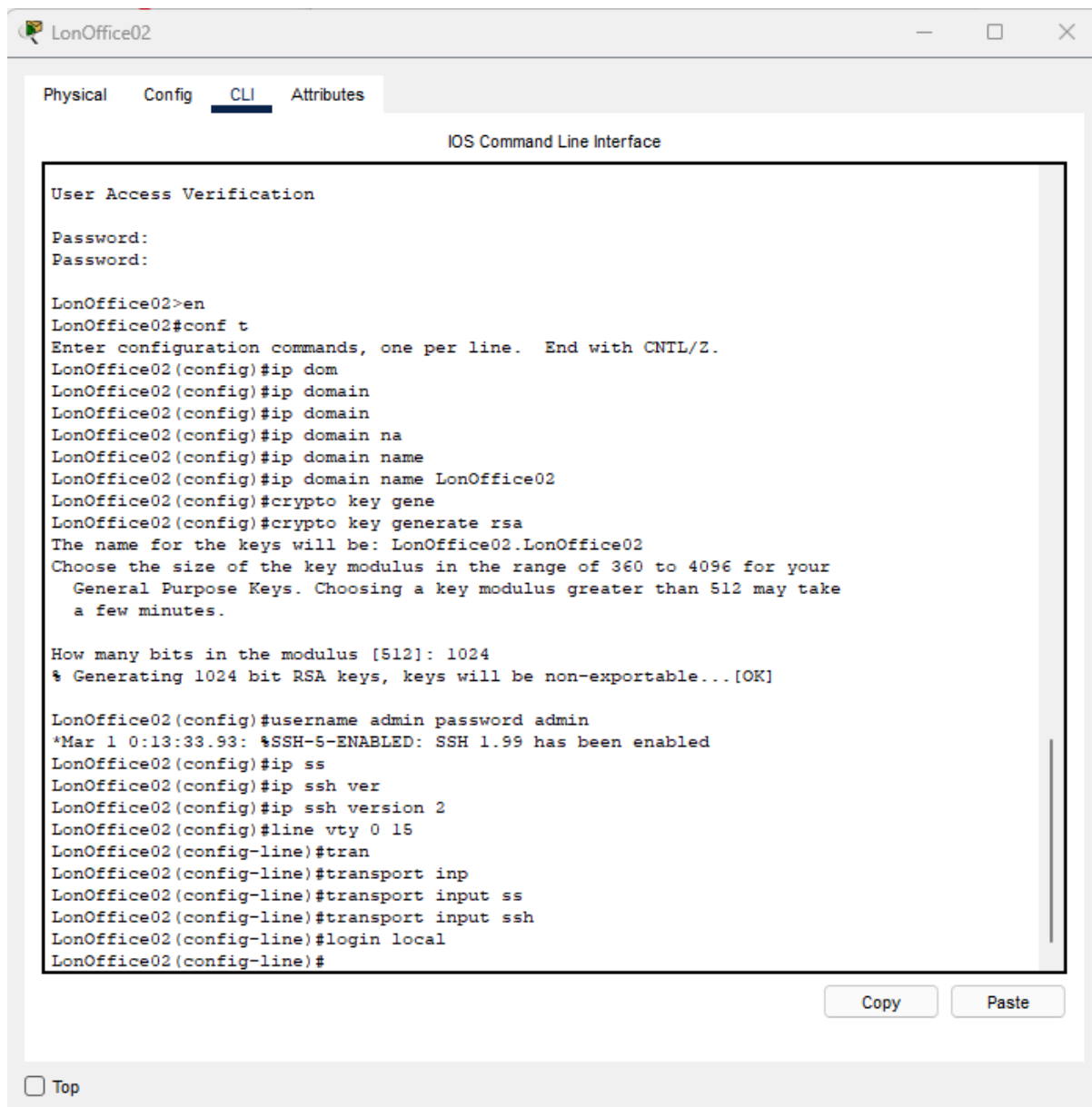
Appendix A: Configuration of hostname and password for London Office 2.



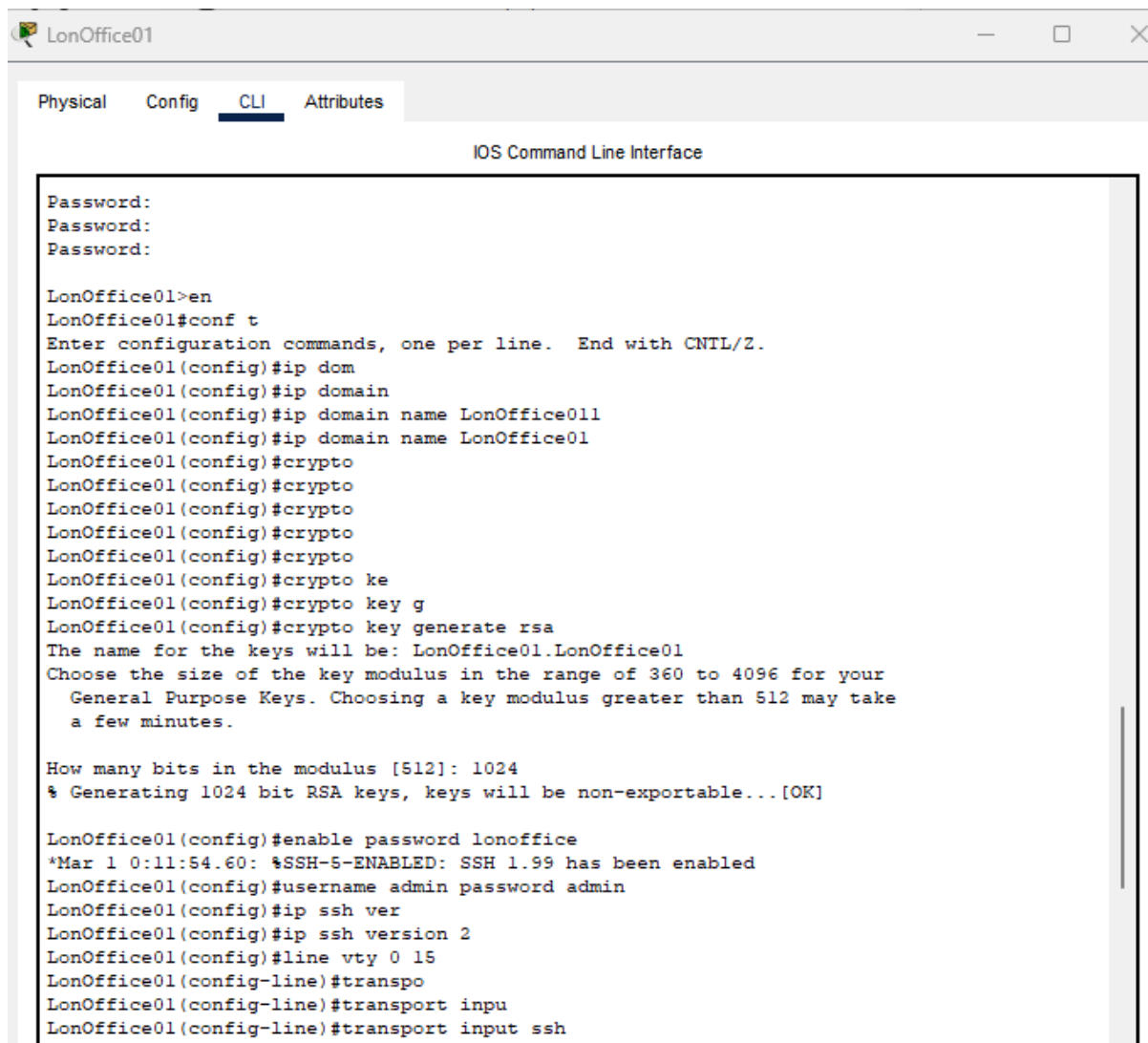
Appendix B: Configuration of hostname and password for Manchester Office 2.



Appendix C: SSH configuration for Manchester Office 2.



Appendix D: SSH configuration for London Office 2.



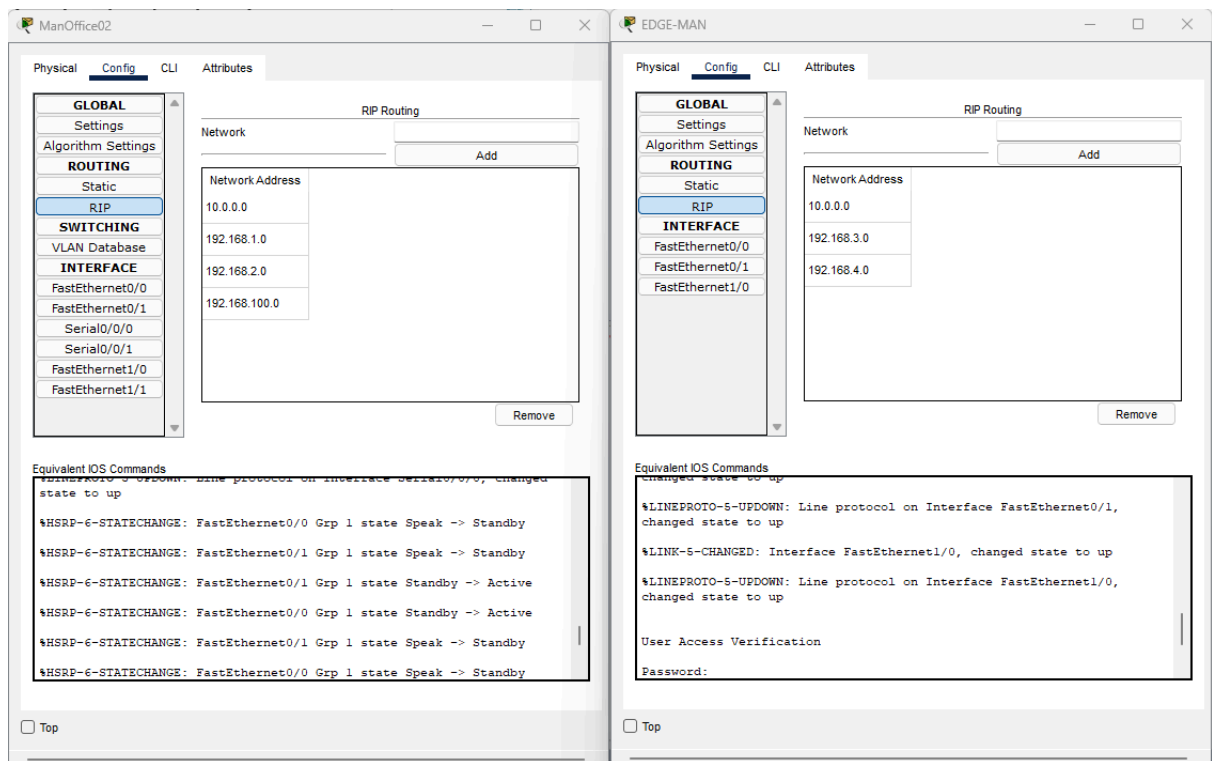
The screenshot shows a window titled 'LonOffice01' with tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The terminal output shows the following sequence of commands and responses:

```
LonOffice01>en
LonOffice01#conf t
Enter configuration commands, one per line. End with CNTL/Z.
LonOffice01(config)#ip dom
LonOffice01(config)#ip domain
LonOffice01(config)#ip domain name LonOffice01
LonOffice01(config)#ip domain name LonOffice01
LonOffice01(config)#crypto
LonOffice01(config)#crypto
LonOffice01(config)#crypto
LonOffice01(config)#crypto
LonOffice01(config)#crypto ke
LonOffice01(config)#crypto key g
LonOffice01(config)#crypto key generate rsa
The name for the keys will be: LonOffice01.LonOffice01
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

LonOffice01(config)#enable password lonoffice
*Mar 1 0:11:54.60: %SSH-5-ENABLED: SSH 1.99 has been enabled
LonOffice01(config)#username admin password admin
LonOffice01(config)#ip ssh ver
LonOffice01(config)#ip ssh version 2
LonOffice01(config)#line vty 0 15
LonOffice01(config-line)#transpo
LonOffice01(config-line)#transport inpu
LonOffice01(config-line)#transport input ssh
```

Appendix E: SSH configuration for London Office 1.



Appendix F: RIP routing configuration for other Manchester Office and building EDGE router.